

3DPS Scene Representation



Optimised 3DPS points (positions, normals, materials)

Forward Model

Antenna Gain

$$G_{TX}(\theta_i) G_{RX}(\theta_o)$$

BSDF

$$f_r(\theta_i, \theta_o, \mathbf{n}_i, \varepsilon'_r, \sigma, t)$$

Phase

$$\varphi_i = -2\pi (R_i^{TX} + R_i^{RX})/\lambda$$

Splat

$$z_i = \sqrt{G_{TX} G_{RX}} f_r e^{j\varphi_i} / R_i^2$$

$$\rightarrow \text{bin } k_i = \text{round}(R_i / \Delta r)$$

Rendered



$$\uparrow \downarrow \mathcal{F}_\theta$$



RA Loss

Ground Truth



$$\uparrow \downarrow \mathcal{F}_\theta$$

